

Necessary in Its Native Block: The Multiplicative Gate, Not Input Selectivity, Underlies Deep-Noise Robustness of a Mamba-3 Block under Matched-Noise Cross-Condition Drift

Yi Wang*

College of Engineering, Lishui University, Lishui 323000, Zhejiang, China

Abstract

The robustness of Mamba-style selective state-space models (SSMs) is often attributed to input-dependent selectivity. We test this attribution with a three-step ablation chain on one controlled testbed: a Mamba-3 block, bearing vibration signals (XJTU-SY, one speed-and-load transition run in both directions), and noise-aware training at matched signal-to-noise ratio. Decision thresholds for each step were written down before the corresponding grid was run; for the last grids they were first committed to a public repository. Values are macro-F1 at the final epoch of a fixed 50-epoch budget, with best-epoch values as a sensitivity analysis. First, replacing the input-dependent parts of Δ , A , B and C by learned constants costs at most about 4 pp under deep noise (paired intervals include zero). Freezing the remaining input-dependent terms of the Mamba-3 scan as well, so that no tensor entering the scan depends on the input, leaves an advantage over a parameter-matched time-invariant S4D of +21.9, +24.0 and +21.2 pp at 0, -2 and -6 dB (5/5 seeds, intervals excluding zero), with fewer parameters. Second, removing the multiplicative SiLU gate from the frozen block costs 9–23 pp and returns it to roughly the level of S4D. Third, replacing the gate in place by an additive branch with identical parameters and initial weights rescues none of that loss: the additive and gateless arms coincide at every level in both directions (differences within ± 0.6 pp, all intervals including

*Corresponding author

Email address: jefflsxy@lsu.edu.cn (Yi Wang)

zero; rescue ratios 0.02 and 0.01, pre-specified verdict: not rescued). Adding a gate to S4D, by contrast, recovers no more than an additive branch of the same size on the original transition (30% versus 41% of the gap) and only a small, direction-dependent increment on the reverse transition. The input-independent scan’s advantage and the cost of gate removal replicate on the reverse transition (+22 to +26 pp and 18–24 pp). Within this block the gate is necessary as a multiplicative operator, and what a time-invariant host gains from an added gate is largely the added structure. A separate per-class grid shows the gate-related differences concentrating on the minority inner-race class. The evidence is conditional on one block, one pair of operating conditions run in both directions with the same eight bearings, five seeds and matched-noise training.

Keywords: state-space models, Mamba, input selectivity, gating, robustness, ablation, distribution shift, matched-noise training

1. Introduction

Selective state-space models (SSMs) of the Mamba family have rapidly become default sequence backbones, and a common narrative has formed around *why* they work: input-dependent selectivity—the ability to modulate state dynamics per token—is credited with filtering noise, adapting to distribution shift, and outperforming time-invariant predecessors such as S4D [1]. [2, 3] This belief motivates a growing family of “selective” architectures, yet it sits in unresolved tension with recent evidence: the first systematic S4D-versus-Mamba comparison on time-series classification finds the *simpler, time-invariant* model consistently superior [4], and theoretical analyses identify expressivity regimes where Mamba is provably weaker [5]. Which account is right—and, more importantly, *what exactly should be credited* when a Mamba-style block does prove robust? This paper addresses the second question for one block and one testbed, with a controlled ablation chain whose decision thresholds were fixed before each grid was run. The testbed is bearing vibration diagnosis under operating-condition drift and matched noise, in which the robustness gap is large (up to 18 pp), the evaluation labels are identical across all 865 cells (hash-verified), and the block can be dissected.

Our approach is deliberately *attributional* rather than *methodological*: we propose no new architecture and claim no state-of-the-art. Instead we take

22 a regime in which a Mamba-style block exhibits a large, stable robustness
 23 advantage over a time-invariant SSM, and ask which ingredient earns it.
 24 The chain has three steps, each with adjudication thresholds fixed before the
 25 corresponding grid was run:

- 26 1. **Test the classical selectivity terms (necessity test).** We con-
 27 struct a *frozen* arm: the input-dependent parts of Δ , A , B and C
 28 are replaced by learned constants. The Mamba-3 scan also takes an
 29 input-dependent trapezoidal weight and input-dependent rotation an-
 30 gles; these, the convolutional stem, the multiplicative SiLU gate, the s-
 31 tate parameterisation, normalisation and projections are retained (Sec-
 32 tion 3.2). If these four terms drive robustness, the frozen arm should
 33 collapse toward the time-invariant baseline.
- 34 2. **Localise the necessary component.** On the frozen base, we remove
 35 one component at a time (convolutional stem; SiLU gate; heavy-tailed
 36 state parameterisation) under a single-variable discipline checked key-
 37 by-key on the state dictionary (Section 3.2), and adjudicate each arm
 38 against pre-specified Δ -thresholds.
- 39 3. **Test sufficiency by constructive grafting.** The component iden-
 40 tified as necessary is added to plain S4D as a gate branch, and a pre-
 41 specified recovery ratio decides whether “removal collapses” is matched
 42 by “addition restores”.

43 The outcome is a conditional pattern (final-epoch reporting; best-epoch
 44 values in Section 4.8). Freezing the input-dependent Δ , A , B , C projections
 45 costs at most about 4 pp in deep noise (paired intervals include zero) while the
 46 block retains a 15–20 pp advantage over S4D: under the pre-specified criterion
 47 these projections are not the primary contributor. Freezing the trapezoidal
 48 weight and the rotation angles as well—so that nothing entering the scan
 49 depends on the input—does not shrink the advantage either: it measures
 50 +21 to +24 pp over S4D with fewer parameters (Section 4.2). Removing the
 51 SiLU gate costs 9–23 pp and takes the block to roughly the level of plain
 52 S4D. Adding a gate branch to S4D recovers only $R = 0.20$ of the gap (0.17
 53 under best-epoch reporting). Rebuilt in the donor’s width and projection
 54 structure, the graft recovers $R = 0.30$ (interval 0.16–0.52), but an additive
 55 branch with identical parameters recovers at least as much (0.41, Section 4.5).
 56 The converse test closes the argument: replacing the gate inside the Mamba-
 57 3 block by an additive branch with identical parameters and initial weights

rescues none of the loss on either transition (rescue ratios 0.02 and 0.01, Section 4.6). The gate is necessary as a multiplicative operator in its native block; in the S4D host it adds nothing beyond an additive branch of the same size on the original transition and a small increment on the reverse one. The first two steps replicate on the reverse transition (Section 4.7). A per-class analysis on a separate, shorter grid indicates that the gate-related differences concentrate on the minority class.

Contributions. (i) A layered freezing experiment on a Mamba-3 block under distribution drift and matched noise, from the classical selectivity terms (Δ , A , B , C) through to a scan in which no tensor depends on the input, in which the block’s deep-noise advantage over S4D survives every step; each arm is verified input-independent by capturing the tensors the scan receives for two different inputs. (ii) Component-level evidence that the output gate is the one ablated component whose removal causes a large, one-signed loss, and that adding it to a time-invariant host, even in the donor’s width and with an additive control of identical size, recovers no more than the added structure does, while replacing it in its own block by an additive branch of identical size recovers nothing on either transition; the first two findings also replicate on the reverse transition. (iii) Final-epoch reporting under a fixed budget for every arm of the chain, with best-epoch selection on the evaluation condition as a sensitivity analysis: absolute levels shift by 0.6–10.6 pp while no contrast changes sign. (iv) A documented pre-specification and review trail—evaluation-label hashes, thresholds written before each grid, and agent-assisted review—including the incidents in which automated adjudication departed from the pre-specified criteria; we release the trail.

2. Background and Related Work

2.1. Time-invariant and selective SSMs

Diagonal structured SSMs (S4D) [6] evolve a linear state with input-independent parameters (A , B , C , Δ), acting as a bank of learned linear filters. Mamba-style blocks [1] make Δ , B , C functions of the input token (selectivity), wrap the scan in a block containing a short causal convolutional stem, a multiplicative SiLU gate on a parallel branch, and residual/normalisation structure; the concrete block used throughout this paper follows the Mamba-3 formulation [7]. The narrative crediting selectivity for robustness draws on theory showing input-dependent dynamics can represent

discontinuity-sensitive projections and counteract memory decay [8], and on [9]. We treat these as *hypotheses to be tested*, not established attributions: the theoretical results concern representability, which does not by itself determine which component earns empirical robustness in a trained block.

2.2. The unresolved comparison

Empirically the picture is contested. A systematic 59-dataset comparison finds S4D-based models superior to Mamba variants on clean time-series classification [4]; theoretical work identifies long-range regimes where Mamba is weaker than S4D [5]. Meanwhile, application literatures—including machine-condition monitoring—contain many Mamba-hybrid methods reporting noise robustness [10, 11, 12, 13, 14], but none, to our knowledge, isolates *which component* of the block is responsible under controlled conditions. Our clean-data results agree with the S4D-superiority finding (S4D matches or slightly exceeds the Mamba-style block on clean data). Whether the disagreement in the literature reflects a regime difference is not settled by a single testbed; what we can report is that, in the drift $\times$ noise regime studied here, the Mamba-style block’s advantage over S4D is not removed by freezing the classical selectivity terms.

2.3. Testbed rationale

We use bearing vibration diagnosis under cross-condition drift (speed and load both change) as the testbed, for three properties that matter to the attribution question rather than to the application: (a) the distribution shift is physically generated (a change of rotation speed and radial load between recorded conditions), not simulated; (b) fault signatures are sparse transients—precisely the discontinuous structures selectivity theory predicts selective SSMs should excel at [8], which makes it a plausible arena for the selectivity hypothesis; and (c) the robustness gap between the Mamba-style block and S4D is large (up to 20 pp), giving the attribution chain high signal. The testbed is two-class (details in Section 3.1); the scope of all claims is bounded accordingly (Section 6).

3. Experimental Protocol

3.1. Testbed, arms, and budget

Vibration windows from the XJTU-SY run-to-failure dataset [15] are split by operating condition: models train on one operating condition (37.5 Hz,

11 kN) and evaluate on another (40 Hz, 10 kN), so speed and load both change (cross-condition), with additive white Gaussian noise applied at matched SNR to training and evaluation ($\{\text{clean}, +10, +6, 0, -2, -6, -10 \text{ dB}\}$). The task is binary fault classification (inner-race vs. outer-race dominant classes; class imbalance makes IR the hard class, Section 4.9). All arms share an identical training budget (50 epochs, 5 seeds; the per-class grid of Section 4.9 used 20 epochs), identical splits, and identical evaluation labels, verified by a hash of the label vector: every reported cell carries `eval_sha = 6c20b367...` (Appendix A), recomputed independently during adjudication. The hash certifies the label vector, not the input windows. In total the study comprises 865 cells across five pre-specified grids (baseline grid 475; frozen-selectivity 105; component ablation 175; graft 35; per-class 75), and four follow-up grids add 255 cells (Sections 4.2, 4.8, 4.5, 4.6 and 4.7).

Composition of the data. Training uses the bearings of the 37.5 Hz/11 kN condition whose failure mode is inner race or outer race (Bearing2_1 is labelled IR; Bearing2_2, 2.4 and 2.5 are labelled OR); evaluation uses those of the 40 Hz/10 kN condition (Bearing3_3 and 3.4 are IR; Bearing3_1 and 3.5 are OR). Bearings with cage or mixed failure modes are excluded. Windows are non-overlapping segments of 2048 samples of the horizontal channel taken from the fault phase, which is delimited by a kurtosis/RMS onset criterion with a last-15% fallback; the evaluation set holds 3,504 OR and 1,744 IR windows, the training set 4,624 OR and 624 IR windows. Two consequences matter for interpretation. The IR class is learned from a single training bearing, so class-level results may reflect bearing-specific generalisation. And the five seeds resample initialisation, batch order and noise but not bearings: there are four independent bearings on each side, and none of the intervals reported here captures bearing-level uncertainty. One follow-up grid (Section 4.7) reverses the transition (40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN). It uses the same eight bearings with their roles swapped, so it tests the other direction rather than independent bearings; its evaluation set holds 4,624 OR and 624 IR windows, all IR windows coming from Bearing2_1, and its label hash is 93424834.... The 35 Hz/12 kN condition has no inner-race bearing and cannot form a binary pair.

The principal arms are summarised in Table 1.

3.2. What the frozen arm freezes, and what the checks verify

In the Mamba-3 layer used here, one input projection emits the gate and value streams, the scan inputs B and C , the step-size and decay terms (Δ

Table 1: Principal experimental arms and parameter counts.

Arm	Description	Params
BM3	Mamba-style block stack (conv stem $k=7 \rightarrow$ selective scan $\rightarrow$ SiLU gate $\rightarrow$ projection), with and without an auxiliary physics loss (Section 3.4)	177,938
BM3-frozen	BM3 with the input-dependent parts of Δ , A , B , C replaced by learned constants; trapezoidal weight, rotation angles, stem, gate and the remaining components unchanged (Section 3.2)	112,410
S4D	Parameter-matched diagonal time-invariant SSM (LayerNorm; no BatchNorm anywhere in either SSM arm)	100,162
S4D-wide	S4D widened to $1.75\times$ parameters (capacity control)	178,498
frozen-conv	Single-component removal on the frozen base: conv stem (Section 4.3)	112,026
frozen-gate	Single-component removal on the frozen base: SiLU gate (Section 4.3)	79,642
frozen-A	Single-component removal on the frozen base: heavy-tailed A parameterisation (Section 4.3)	112,410
frozen-trap	Frozen base with the trapezoidal weight also made a learned constant (Section 3.2)	111,906
frozen-rot	Frozen base with the rotation angles also made learned constants	104,346
frozen-all	Frozen base with both also constant: no tensor entering the scan depends on the input	103,842
S4D+gate	Plain S4D plus the SiLU gate branch only (graft, Section 4.4)	116,802
S4D+gate (wm)	Graft rebuilt in the donor’s shape: per layer x, z projected to width 128, S4D at width 128 (state 64, same SSM budget as the host), $W_{\text{out}}(y \odot \text{SiLU}(z))$ (Section 4.5)	198,978
S4D+branch	Same modules, parameters and initial weights, combined additively: $W_{\text{out}}(y + \text{SiLU}(z))$ (control)	198,978
frozen+add	BM3-frozen with the gate replaced in place: $W_{\text{out}}(y + \text{SiLU}(z))$ instead of $W_{\text{out}}(y \odot \text{SiLU}(z))$; same modules ⁷ parameters and initial weights (Section 4.6)	112,410
CNN/TCN baselines	Conventional baselines for context	—

165 and A , through input-dependent additive terms), a trapezoidal weight and
 166 rotation angles. The frozen arm removes the input-dependent parts of Δ ,
 167 A , B and C and replaces them with learned constants broadcast over time
 168 (for Δ the static bias of the original layer is kept; for A a new constant pa-
 169 rameter plays the role of a static log-decay, passed through the layer’s own
 170 heavy-tailed clamp). The removed projections are deleted, so no parameter
 171 or gradient can leak. **The trapezoidal weight and the rotation angles**
 172 **remain functions of the current input**, as do the stem and the output
 173 gate. The frozen arm is therefore not a time-invariant scan: in the Mamba-3
 174 formulation [7] the input-dependent rotation and trapezoidal weighting also
 175 enter the state update. Section 4.1 is accordingly a statement about the
 176 classical selectivity terms Δ, A, B, C only. Three further arms close that gap
 177 by the same device: *frozen-trap* replaces the trapezoidal weight by a learned
 178 per-head constant, *frozen-rot* replaces the rotation angles by learned con-
 179 stants (so the rotation becomes a learned position-only phase rather than a
 180 function of the current input), and *frozen-all* does both, leaving **no tensor**
 181 **entering the scan dependent on the input**. In each case the correspond-
 182 ing `in_proj` output slice is deleted. Section 4.2 reports them. For these arms
 183 the construction check adds the decisive probe: the tensors handed to the
 184 scan are captured for two different inputs through the same module, and a
 185 tensor the arm claims to have frozen is required to be bitwise identical across
 186 them while a tensor left selective is required to differ (all three arms pass).
 187 No kernel modification is needed, because the scan interface accepts these
 188 tensors regardless of their origin.

189 Single-variable discipline was checked by key-level state-dictionary diffs
 190 between two models built under the same seed. For the frozen arm the
 191 diff verifies that every non-target tensor has the same name and shape as in
 192 BM3; for the component-ablation and graft arms it additionally compares the
 193 values of non-target tensors at initialisation. Every differing key is mapped
 194 to an entry in an architecture map. We do not claim byte-level identity of
 195 non-target tensors in the frozen arm, and trained weights naturally differ
 196 between arms; the checks concern how the arms were constructed.

197 3.3. Reporting rule

198 Every cell is trained for a fixed budget of 50 epochs with a cosine learning-
 199 rate schedule that anneals to zero, and the macro-F1 reported in the main
 200 text is that of the *final epoch*, a rule that uses no evaluation data to choose
 201 a model. The original grids recorded instead the maximum over epochs of

macro-F1 on the evaluation split (the harness defines no separate validation split), and the thresholds pre-specified for those grids were written against that rule. Every arm of the chain was therefore re-run on the adjudication band (0, -2, -6 dB) with the original training code, the same evaluation fingerprint and per-epoch evaluation (Section 4.8). Adjudications are reported under the rule they were specified for and under the final-epoch rule, and best-epoch values are given as a sensitivity analysis. Levels outside the band (clean, +10, +6, -10 dB) and the per-class grid were not re-run; values there are best-epoch and labelled auxiliary. Because the final epoch of the same evaluation condition has been inspected repeatedly during this study, final-epoch values are not a fresh confirmatory test either.

3.4. Confound elimination

Before component attribution, three candidate confounds were examined on the same grid. **Capacity:** S4D-wide ($1.75\times$ parameters) tracks S4D (S4D-wide - S4D: -0.4, -1.0, +0.4 pp at 0, -2, -6 dB under final-epoch reporting, all intervals including zero), while BM3-frozen exceeds S4D-wide by +15.9, +20.8, +16.0 pp with intervals excluding zero at all three levels, so this particular increase in S4D’s parameter count does not explain the gap; the control does not address the change in channel mixing that gate removal also entails. **Normalisation:** both SSM arms use LayerNorm and contain no BatchNorm; the arm that degrades under noise is the one *without* any batch statistics, which argues against BatchNorm statistics mismatch as the explanation. Other normalisation inside the Mamba-3 layer was not varied. **Auxiliary physics loss:** an exploratory kinematics-alignment regulariser carried by the original model contributes -0.1 pp pointwise (curves coincide); all attribution arms are therefore built on the pure cross-entropy variant.

3.5. Pre-specification and adjudication

Every adjudicated question in this study had numeric decision thresholds written to a task card and to disk before the corresponding grid was launched. This is internal pre-specification, not external registration: the supporting evidence is local file modification times and run-directory names (released, Appendix A), which are not tamper-evident and are not preserved by a repository checkout, so the trail lets readers inspect the order of events but cannot prove it. The exception is the last three follow-up grids (Sections 4.5–4.8): their thresholds, and for the last two grids the title decision that depends on their outcome, were committed to the public

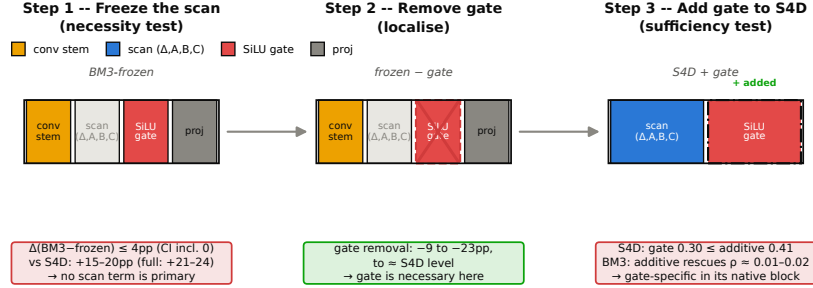
repository before the first cell was run, which gives those grids an external
 time-stamp. Later questions (component ablation, graft) were specified after
 earlier grids had been seen, as is natural in a staged design. Adjudication
 was mechanical: an analysis pass applies the thresholds and emits a decision
 artifact; an independent reviewer pass re-computes aggregates, verifies finger-
 prints row-by-row, and must cite file paths and modification times for every
 check. The pre-specified thresholds were binding in practice: two attempted
 post-hoc criterion substitutions and one stale-artifact false verification were
 caught and re-adjudicated under the original rules — this is the one place
 where pre-specification actually bit. [Appendix C](#) details the incidents, the
 automated executor/reviewer implementation, and the human adjudication
 gates; all scientific decisions — thresholds, rulings, and interpretations —
 were made by the authors and are timestamped. Statistical tests (paired,
 per-condition) are reported throughout but were never adjudication criteria.
 A single repeat of one 2-epoch configuration in a separate process differed
 by ~ 0.7 pp macro-F1 (non-deterministic kernel reductions; [Appendix B](#));
 this is an observed repeat difference, not an estimated variability bound, and
 arm-level contrasts are judged against seed-level variability (2–9 pp).

4. Results

Throughout, “deep noise” denotes the pre-specified adjudication band
 $\{0, -2, -6 \text{ dB}\}$; -10 dB is reported but excluded from adjudication (the time-
 invariant baseline is near chance there: a constant majority-class predictor
 scores 40.0% and a uniformly random predictor 48.6% macro-F1 on this
 evaluation set, against 52.3% for S4D at the best epoch). All values are
 macro-F1 (%), mean over 5 seeds, at the final epoch of the 50-epoch budget
 (Section 3.3); best-epoch values are given alongside or in Section 4.8, and
 values outside the adjudication band are best-epoch and auxiliary. Figure 1
 summarises the three-step ablation chain that structures this section.

4.1. Freezing the input-dependent Δ , A , B , C projections

Freezing the input-dependent Δ , A , B , C projections (Section 3.2) costs
 at most 3.7 pp anywhere in the adjudication band on the point estimates
 under final-epoch reporting (+3.7, +0.9, +0.2 pp at 0, -2 , -6 dB), while
 the frozen block retains an advantage of +15.5, +19.8, +16.4 pp over S4D
 (Table 2, Figure 2). Five seeds bound these quantities loosely: paired 95% in-
 tervals for $\Delta(\text{BM3-frozen})$ are $[-3.8, 11.3]$, $[-4.5, 6.3]$, $[-12.1, 12.6]$ pp (BM3



Conditional on this block, testbed and budget: the gate is necessary as a multiplicative operator in its native block.

Figure 1: The three-step ablation chain with pre-specified thresholds: freeze the input-dependent Δ, A, B, C projections (Section 4.1), remove one component at a time on the frozen base to localise the necessary ingredient (Section 4.3), and graft that ingredient onto S4D to test sufficiency (Section 4.4).

Table 2: BM3, BM3-frozen and S4D, macro-F1 (%), mean over 5 seeds; Δ columns are seed-paired mean differences in percentage points. Upper block: final epoch of the 50-epoch budget on the adjudication band (primary; re-runs of Sections 4.2 and 4.8). Middle block: best epoch on the evaluation condition in the same re-runs (sensitivity). Lower block: best epoch in the original frozen-selectivity grid for levels outside the band (auxiliary; not re-run).

Condition	BM3	BM3-frozen	S4D	$\Delta(\text{BM3}-\text{frozen})$	$\Delta(\text{frozen}-\text{S4D})$
<i>Final epoch (primary)</i>					
0 dB	86.9	83.1	67.6	+3.7	+ 15.5
-2 dB	88.3	87.3	67.5	+0.9	+ 19.8
-6 dB	83.3	83.1	66.7	+0.2	+ 16.4
<i>Best epoch (sensitivity, same re-runs)</i>					
0 dB	95.6	90.6	75.9	+5.1	+14.6
-2 dB	95.8	92.3	75.1	+3.5	+17.2
-6 dB	88.6	85.7	68.6	+3.0	+17.1
<i>Auxiliary (best epoch, original grid)</i>					
clean	78.9	78.9	81.0	-0.0	-2.1
-10 dB*	68.1	64.3	52.3	+3.8	+12.0

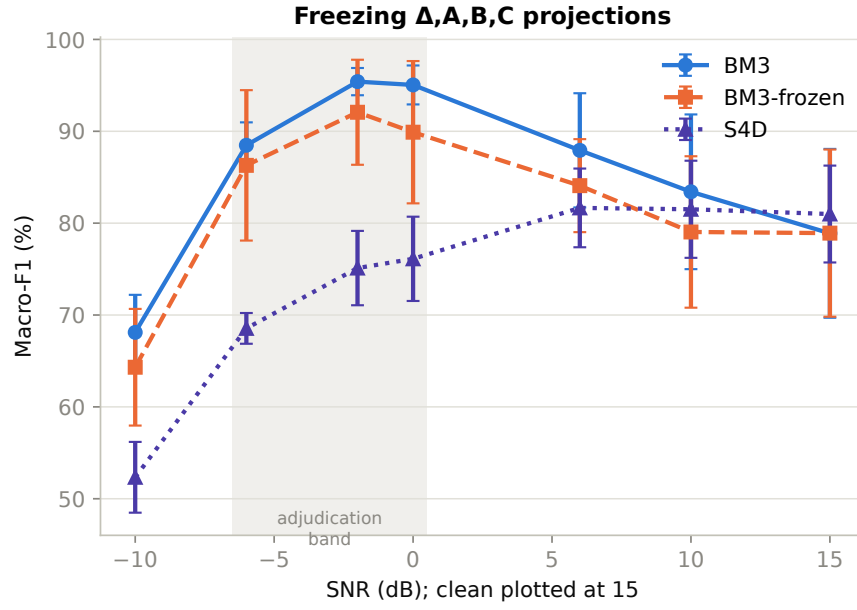


Figure 2: Macro-F1 vs. SNR for BM3, BM3-frozen, and S4D across the full noise grid (best epoch, original frozen-selectivity grid; final-epoch values on the adjudication band are in Table 2, and levels outside the band are auxiliary). The frozen arm tracks BM3 closely and stays well above S4D throughout the adjudication band.

273 ahead in 4/5, 3/5, 2/5 seeds), and for $\Delta(\text{frozen-S4D})$ [4.3, 26.8], [12.0, 27.7],
 274 [4.1, 28.7] pp (5/5, 5/5, 4/5 seeds). The pre-specified rule attributed robust-
 275 ness to selectivity only if $\Delta(\text{BM3-frozen}) \geq 10$ pp in deep noise with the
 276 frozen arm collapsing toward S4D. It was written for best-epoch reporting,
 277 under which the original grid gives $\Delta(\text{BM3-frozen}) = +5.1, +3.3, +2.2$ pp
 278 (re-run: +5.1, +3.5, +3.0) and $\Delta(\text{frozen-S4D}) = +13.8, +17.0, +17.7$ pp.
 279 Neither condition holds under either rule, so the pre-specified verdict is that
 280 these projections are *not the primary mechanism*. This is a failure to meet
 281 a confirmation criterion, not a demonstration that freezing is cheap: the
 282 intervals above are compatible with differences larger than 10 pp, and no e-
 283 quivalence margin was pre-specified or tested. One limit applies: the frozen
 284 arm still contains input-dependent trapezoidal weights and rotation angles, so
 285 this result concerns Δ, A, B, C only; Section 4.2 removes that limit. On clean
 286 data (auxiliary, best epoch) S4D matches or slightly exceeds both Mamba-
 287 style arms—consistent with the systematic clean-TSC comparison [4]. In
 288 the noisy regime, the block’s advantage over S4D is not removed by freezing
 289 these four terms, so it must be carried by other parts of the block or by the
 290 remaining input-dependent scan terms.

291 4.2. Freezing the remaining input-dependent scan terms

292 Table 3 reports the three layered arms of Section 3.2 on the adjudication
 293 band (60 cells: 4 arms $\times$ 3 levels $\times$ 5 seeds $\times$ 50 epochs, the original training
 294 code, the same evaluation fingerprint, with the original BM3 arm re-run
 295 in the same grid). The question is whether the advantage that survived
 296 freezing Δ, A, B, C is carried by the two input-dependent terms the frozen
 297 arm retained.

298 It is not. With every tensor entering the scan input-independent, *frozen-all*
 299 is +21.9, +24.0 and +21.2 pp above S4D at 0, −2 and −6 dB under final-
 300 epoch reporting, positive in 5/5 seeds at every level and with intervals ex-
 301 cluding zero—if anything a larger margin than the partially frozen arm’s
 302 +15.5, +19.8, +16.4 pp (Table 10). None of the three freezing steps costs
 303 accuracy: the paired differences BM3-frozen − frozen-trap, − frozen-rot
 304 and − frozen-all are negative at all three levels (between −1.8 and −6.3 pp
 305 under final-epoch reporting), i.e. the more heavily frozen arm is numerically
 306 ahead, with every interval including zero. The direction is not an artefact of
 307 capacity: *frozen-all* has 103,842 parameters against 112,410 for BM3-frozen
 308 and 177,938 for BM3. At the deepest level *frozen-all* also exceeds the full
 309 BM3 block by 4.5 pp (−7.2, −1.9) for BM3 − frozen-all, 0/5 seeds in BM3’s

Table 3: Layered freezing of the scan’s input-dependent terms. Macro-F1 (%), mean over 5 seeds, under best-epoch and final-epoch reporting; Δ columns are paired differences under final-epoch reporting with 95% t -intervals over seeds.

Arm	0 dB		−2 dB		−6 dB	
	best	final	best	final	best	final
BM3 (re-run)	95.6	86.9	95.8	88.3	88.6	83.3
BM3-frozen (Δ, A, B, C)	90.6	83.1	92.3	87.3	85.7	83.1
frozen-trap	95.5	87.6	95.3	89.3	90.5	88.0
frozen-rot	94.6	86.7	94.6	89.1	89.0	86.4
frozen-all	96.5	89.5	96.8	91.5	90.5	87.9
S4D	75.9	67.6	75.1	67.5	68.6	66.7
<i>Paired differences, final-epoch reporting (pp)</i>						
frozen-all − S4D	+21.9 [16.7, 27.1]		+24.0 [21.4, 26.5]		+21.2 [17.5, 24.9]	
BM3-frozen − frozen-all	−6.3 [−17.6, 4.9]		−4.1 [−12.0, 3.7]		−4.8 [−17.8, 8.2]	
BM3 − frozen-all	−2.6 [−8.2, 2.9]		−3.2 [−7.0, 0.6]		−4.5 [−7.2, −1.9]	

310 favour); we report this but do not build on it, since five seeds and one speed
311 pair cannot separate a genuine advantage of a time-invariant scan from a
312 variance effect.

313 Two readings of the whole step are therefore available, and the data do not
314 separate them: input-dependence of the scan contributes nothing measurable
315 here, or it contributes something that the learned constants recover under
316 this training budget. Either way the conclusion for the attribution chain
317 is the same and now rests on a complete intervention rather than a partial
318 one: **the Mamba-style block’s deep-noise advantage over a time-**
319 **invariant S4D does not require any input-dependent term in its**
320 **scan.** What remains to carry it is the rest of the block—the convolutional
321 stem, the heavy-tailed A parameterisation, the normalisations and the output
322 gate—which is what Sections 4.3–4.4 examine. The re-run also reproduces
323 the original grids: the BM3 arm’s best-epoch means (95.6, 95.8, 88.6) exceed
324 the original frozen-selectivity grid’s 95.0, 95.4, 88.5 by 0.57, 0.43 and 0.16 pp
325 (a different PyTorch/Triton build, Section 4.8).

326 4.3. Removing the gate is the only ablation with a large, one-signed effect

327 “Necessary” is used here in the operational sense of this ablation: remov-
328 ing the gate from the frozen block, with the other components and the budget
329 unchanged, removes most of the block’s advantage. The gate result is large,
330 one-signed and grows with noise depth: under final-epoch reporting removal

Table 4: Single-component removal on the frozen base, $\Delta(\text{frozen} - \text{arm})$ in percentage points: final-epoch mean, 95% t -interval over seeds and number of seeds (of 5) with a positive difference (re-runs of Section 4.8); “best” is the best-epoch value of the original component-ablation grid, for which the thresholds were pre-specified.

Component removed	0 dB	−2 dB	−6 dB	Adjudication
SiLU gate	+9.1 [2.9, 15.3] (5/5); best +13.1	+15.6 [−0.8, 32.0] (4/5); best +21.1	+22.6 [5.8, 39.4] (5/5); best +23.5	PRIMARY (all $\geq$ 8 pp, both rules)
conv stem ($k=7$)	−13.2 [−24.1, −2.2] (0/5); best −6.6	−7.5 [−14.6, −0.3] (0/5); best −2.7	+5.9 [−4.0, 15.8] (3/5); best +7.9	outside bins; regime trade-off
heavy-tailed A → S4D-style A	−1.9 [−9.2, 5.3] (2/5); best −2.4	−2.6 [−8.6, 3.4] (2/5); best −2.2	−4.8 [−16.4, 6.9] (2/5); best −4.2	outside bins; inter- vals include 0

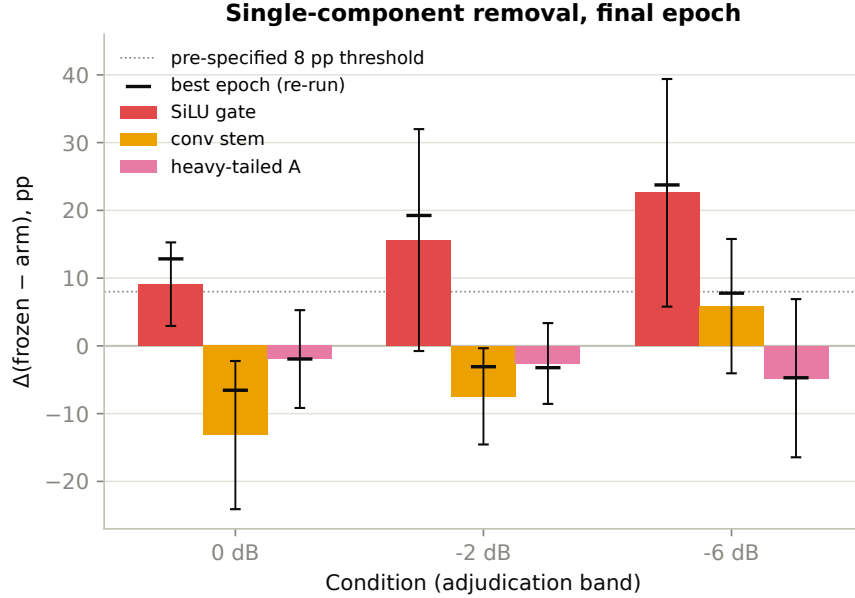


Figure 3: Component-removal Δ (percentage points, frozen $-$ arm) on the adjudication band for the three single-component ablations on the frozen base: final-epoch means with 95% t -intervals over seeds (bars) and best-epoch means of the same re-runs (black ticks). Only gate removal is large and one-signed at every level and clears the pre-specified 8 pp threshold.

costs 9.1, 15.6, 22.6 pp at 0, -2 and -6 dB (paired 95% intervals [2.9, 15.3], $[-0.8, 32.0]$, [5.8, 39.4]; 5/5, 4/5, 5/5 seeds), above the pre-specified 8 pp threshold at every level; the interval at -2 dB includes zero because of the large seed variance of the gateless arm. Under the best-epoch rule for which the threshold was written, the component-ablation grid gives 13.1, 21.1 and 23.5 pp ([6.5, 19.6], [3.8, 38.5], [8.5, 38.6]; 5/5 seeds at every level), so the verdict is *primary* under both rules. The gateless arm gives up most of the Mamba advantage but is not resolved against plain S4D: frozen-gate $-$ S4D is +6.4, +4.2, -6.2 pp, with every interval including zero. We therefore describe it as roughly at the level of S4D, below it only on the point estimate at -6 dB; this suggests, but does not establish, that components interact rather than add. The other two components tell regime stories rather than robustness stories. Removing the convolutional stem is beneficial at 0 and -2 dB (frozen $-$ frozen-conv = -13.2 [$-24.1, -2.2$] and -7.5 [$-14.6, -0.3$], with the stemless arm ahead in every seed) and costly at -6 dB (+5.9, [$-4.0, 15.8$], 3/5 seeds); under best-epoch reporting the pattern was the same ($-6.6, -2.7, +7.9$ pp). On clean cross-speed data (auxiliary, best epoch, original grid) the stem costs 21.0 pp (removal yields 98.8 vs. 77.8, with seed-std collapsing from 8.1 to 0.6 pp)—consistent either with speed-specific local temporal patterns mistransferring across conditions, or with optimisation friction under the matched budget; we report both readings. Replacing the heavy-tailed state parameterisation by an S4D-style one is numerically beneficial at all three levels (by 1.9, 2.6, 4.8 pp under final-epoch and 2.4, 2.2, 4.2 pp under best-epoch reporting), but every paired interval includes zero, so we draw no directional conclusion from it.

4.4. Adding a narrow gate branch to S4D

The pre-specified recovery ratio is $R = \text{mean over the band of (S4D+gate-S4D)/(frozen-S4D)}$. Under final-epoch reporting $R = \mathbf{0.196}$ (per level 0.249, 0.341, -0.003 ; seed-resampled 95% interval [0.09, 0.31]); under the best-epoch rule for which it was specified, $R = 0.174$ ([0.04, 0.29]). The pre-specified bins were $R \geq 0.7$ (confirmed), $0.3 \leq R < 0.7$ (partial recovery) and $R < 0.3$ (insufficient); both values fall in the last, although the final-epoch interval reaches the 0.3 boundary. The per-level paired gains of S4D+gate over S4D under final-epoch reporting are +3.9 ($[-0.8, 8.6]$), +6.8 ([2.4, 11.1]), -0.1 ($[-4.0, 3.9]$) pp. At the deepest adjudicated level the graft recovers nothing. Removal and addition are not mirror-image interventions. In BM3 the gate has its own $64 \rightarrow 128$ projection and multiplies the scan

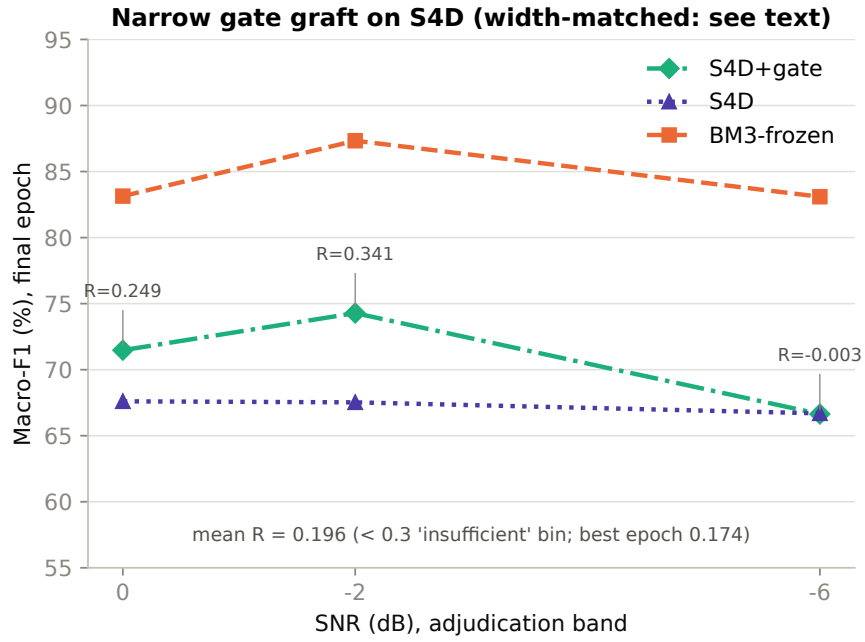


Figure 4: Graft outcome under final-epoch reporting: S4D+gate, plain S4D, and BM3-frozen on the adjudication band (one re-run grid), with the per-level recovery ratio $R = (S4D+gate - S4D) / (frozen - S4D)$ annotated. Recovery is near zero at the deepest adjudicated noise level. This is the narrow graft; the width-matched graft and its additive control are in Table 6.

Table 5: Grafting the SiLU gate branch onto plain S4D (S4D unchanged key-by-key; only the gate branch added). Upper block: final epoch, all three arms from the same re-run (Section 4.8). Lower block: best epoch as pre-specified, with S4D+gate from the graft grid and S4D and BM3-frozen, and hence R , from the component-ablation grid of Section 4.3 (independent runs of the same arms differ by 0.1–0.7 pp).

Condition	S4D+gate	S4D	BM3-frozen	Recovery ratio
<i>Final epoch (primary)</i>				
0 dB	71.5	67.6	83.1	0.249
−2 dB	74.3	67.5	87.3	0.341
−6 dB	66.6	66.7	83.1	−0.003
<i>Best epoch (as pre-specified)</i>				
0 dB	79.8	76.2	90.6	0.249
−2 dB	79.4	75.2	92.8	0.239
−6 dB	69.2	68.6	86.0	0.035

output at the inner width of 128 before the output projection; removing it deletes 32,768 parameters. In the S4D host the grafted gate is a $64 \rightarrow 64$ linear branch (with bias) computed from the layer input and multiplying the SSM output before the residual addition; it adds 16,640 parameters over four layers. The two share the multiplicative SiLU form but not width, projection structure or parameter count. Removal costs 9–23 pp in the frozen block, whereas this graft changes S4D by between −0.1 and +6.8 pp (final epoch). What this shows is that *this gate branch, in this host and under this budget, recovers little*. Three readings fit the data and cannot be separated here: (i) host dependence, i.e. the gate needs the stem or scan context it co-trained with; (ii) implementation differences in width, projection and placement; (iii) transplant friction, i.e. the grafted branch is harder to optimise in a foreign host under the matched budget (training converges normally, but convergence is not robustness). Section 4.5 tests reading (ii) with a width-matched graft and an equal-capacity non-multiplicative control; a tuning or longer-budget comparison selected on a source-domain validation set was not run.

The four arms do allow a coarse host-by-gate contrast, $I = [F(\text{frozen}) - F(\text{frozen-gate})] - [F(\text{S4D+gate}) - F(\text{S4D})]$, where F is macro-F1. Under last-epoch reporting $I = 5.2$ [2.3, 8.1], 8.9 [−6.2, 23.9] and 22.6 [2.9, 42.4] pp at 0, −2 and −6 dB (seed-matched 95% t -intervals; positive in 5/5, 4/5 and 5/5 seeds); under best-epoch reporting in the re-run it is 7.8 [1.4, 14.2], 14.4

389 $[-1.4, 30.3]$ and 23.1 $[8.7, 37.4]$ pp. A positive I means the gate adds more
390 on the frozen host than on S4D. Because the host differs in several respects
391 at once and the two gates are not aligned, I cannot be assigned to a specific
392 component or interaction term, and the seed pairing across arms is nominal
393 (arms share seed indices, not initialisations).

394 4.5. A width-matched graft and an additive control

395 The graft of Section 4.4 is narrower and smaller than the donor gate.
396 Two further arms rebuild it in the donor’s shape (Table 1). In each layer
397 the layer input is projected to x and z of width 128 (the frozen block’s
398 inner width), the S4D layer runs on x at width 128 with state size 64—so
399 that its own parameter count and total state size equal the host’s 64-wide,
400 state-128 layer—and an output projection maps back to width 64. *S4D+gate*
401 (*wm*) combines $y \odot \text{SiLU}(z)$, as the donor does; *S4D+branch* uses the same
402 modules, parameters and initial weights but combines $y + \text{SiLU}(z)$. Both have
403 198,978 parameters, more than BM3-frozen (112,410) and BM3 (177,938),
404 which biases the test towards recovery. A construction check with forward
405 hooks confirms that the tensor entering the output projection equals the
406 intended formula for each arm and not the other, and that the two arms are
407 identical at initialisation. The thresholds were committed publicly before
408 the grid ran (Section 3.5): R as before, under final-epoch reporting; the
409 verdict is INCONCLUSIVE whenever the seed-resampled interval crosses a
410 bin boundary, in which case we write at the tier less favourable to the claim;
411 and a gate-specific reading requires $\Delta R = R_{\text{gate}} - R_{\text{branch}} > 0.1$ with an
412 interval excluding zero.

Table 6: Width-matched graft and additive control on the original transition. Macro-F1
(%), mean over 5 seeds, final-epoch reporting; R is the mean over the three levels of (arm
– S4D)/(BM3-frozen – S4D), with a seed-resampled 95% interval. S4D, BM3-frozen and
the narrow graft are the re-runs of Section 4.8.

Arm	0 dB	–2 dB	–6 dB	R	95% interval
S4D	67.6	67.5	66.7	–	–
S4D+gate (narrow)	71.5	74.3	66.6	0.196	[0.09, 0.31]
S4D+gate (wm)	70.1	73.4	73.8	0.296	[0.16, 0.52]
S4D+branch	73.1	76.2	73.8	0.408	[0.31, 0.60]
BM3-frozen	83.1	87.3	83.1	1	–
$\Delta R = \text{S4D+gate (wm)} - \text{S4D+branch}$				–0.112	[–0.26, 0.02]

413 The width-matched graft recovers $R = 0.296$ of the gap (per level 0.159,
 414 0.298, 0.431). Its interval, $[0.16, 0.52]$, crosses the 0.3 boundary, so the pre-
 415 specified verdict is INCONCLUSIVE between “insufficient” and “partial”,
 416 and we report it at the less favourable tier: a graft built in the donor’s shape
 417 recovers part of the gap. The recovery is not gate-specific. The additive
 418 branch, with the same parameters, recovers at least as much ($R = 0.408$),
 419 $\Delta R = -0.112$ $[-0.26, 0.02]$, and at 0 dB the gated arm is below the additive
 420 one by 3.0 pp ($[-4.5, -1.5]$, 0/5 seeds). With frozen—all in the denominator
 421 the gated arm’s ratio is 0.231 $[0.13, 0.33]$; under best-epoch reporting the two
 422 ratios are 0.459 and 0.483. What a graft recovers on this transition there-
 423 fore follows the added structure—a wider channel axis and input/output
 424 projections—rather than the multiplicative interaction. Reading (ii) of Sec-
 425 tion 4.4 is partly confirmed, in that the narrow graft understated what added
 426 structure can recover; but in the S4D host the gate itself contributes noth-
 427 ing beyond an equal-capacity additive branch, whereas removing it from the
 428 frozen Mamba-3 block costs 9–23 pp under the same rule (Table 10).

429 4.6. Replacing the gate in its native host

430 The previous subsection changes the host and keeps the operator. A pre-
 431 specified grid (15 cells, original transition) does the converse: it keeps the host
 432 and changes the operator. In *frozen+add* the scan kernel of every BM3-frozen
 433 layer is called without gating and the layer output becomes $W_{\text{out}}(y + \text{SiLU}(z))$;
 434 modules, parameter count (112,410) and initial weights are bitwise those of
 435 BM3-frozen, and a construction check confirms that the donor operator is
 436 $y \odot \text{SiLU}(z)$ and that the new arm computes $y + \text{SiLU}(z)$. The pre-specified
 437 statistic is the native rescue ratio $\rho = (\text{add} - \text{no gate})/(\text{frozen} - \text{no gate})$,
 438 mean over levels, final epoch, with a seed-resampled interval; *not rescued*
 439 requires $\rho < 0.3$ with an upper bound below 0.5.

440 The additive branch rescues nothing. *frozen+add* is indistinguishable
 441 from the gateless arm at every level (+0.6, −0.2, +0.4 pp, every interval
 442 including zero), and the multiplicative arm exceeds it by 8.5, 15.8, 22.2 pp.
 443 The rescue ratio is $\rho = 0.024$ (per level 0.070, −0.015, 0.016), with a seed-
 444 resampled 95% interval of $[-1.159, 0.486]$; under best-epoch reporting $\rho =$
 445 0.005. By the pre-specified rule the verdict is *not rescued*. Two features of
 446 the interval should be read with it: its upper bound lies only 0.014 below the
 447 threshold, and its lower bound is wide because the 0 dB denominator is only
 448 9.1 pp (in 7.8% of resamples one level’s denominator fell below 2 pp and was
 449 excluded, as pre-specified).

Table 7: Gate replaced by an additive branch inside BM3-frozen, original and reverse transitions. Macro-F1 (%), mean over 5 seeds; paired differences under final-epoch reporting with 95% t -intervals and the number of seeds (of 5) with a positive difference.

Arm	0 dB		−2 dB		−6 dB	
	best	final	best	final	best	final
<i>Original transition (37.5 Hz/11 kN → 40 Hz/10 kN)</i>						
BM3-frozen ($y \odot \text{SiLU}(z)$)	90.6	83.1	92.3	87.3	85.7	83.1
frozen+add ($y + \text{SiLU}(z)$)	78.0	74.7	72.6	71.5	62.3	60.9
frozen−gate (no z)	77.7	74.0	73.1	71.7	61.9	60.5
BM3-frozen − frozen+add	+8.5		+15.8		+22.2	
	[1.1, 15.9] (5/5)		[−1.2, 32.9] (4/5)		[4.6, 39.9] (5/5)	
frozen+add − frozen−gate	+0.6		−0.2		+0.4	
	[−1.7, 3.0] (3/5)		[−3.4, 3.0] (3/5)		[−0.7, 1.4] (3/5)	
<i>Reverse transition (40 Hz/10 kN → 37.5 Hz/11 kN)</i>						
BM3-frozen ($y \odot \text{SiLU}(z)$)	81.1	74.7	84.0	76.7	80.8	73.9
frozen+add ($y + \text{SiLU}(z)$)	64.3	56.5	62.2	54.4	55.7	50.5
frozen−gate (no z)	63.8	56.2	62.0	54.7	57.1	50.0
BM3-frozen − frozen+add	+18.1		+22.3		+23.4	
	[1.2, 35.0] (5/5)		[3.5, 41.0] (5/5)		[6.8, 40.0] (5/5)	
frozen+add − frozen−gate	+0.3		−0.3		+0.6	
	[−0.6, 1.2] (3/5)		[−1.0, 0.4] (2/5)		[−0.2, 1.4] (5/5)	

450 A further pre-specified grid (15 cells) repeats the substitution on the re-
451 verse transition (Section 4.7), with BM3-frozen and frozen-gate taken from
452 the reverse grid. The outcome is the same (Table 7, lower block): frozen+add
453 coincides with the gateless arm at every level (+0.3, -0.3, +0.6 pp, every in-
454 terval including zero), and the multiplicative arm exceeds it by +18.1, +22.3,
455 +23.4 pp (intervals excluding zero, 5/5 seeds at every level). The rescue ra-
456 tio is $\rho = 0.009$ (per level 0.017, -0.014, 0.023), with a seed-resampled 95%
457 interval of $[-0.943, 0.362]$; under best-epoch reporting $\rho = -0.005$. The pre-
458 specified verdict is again *not rescued*, here with the upper bound 0.14 below
459 the threshold.

460 Together with Section 4.5 this gives the two halves of a symmetric test.
461 In the S4D host the gate adds nothing beyond an additive branch of the same
462 size; in the Mamba-3 host an additive branch of the same size adds nothing
463 beyond removing the gate, on both transitions. The benefit of the gate is
464 specific to the multiplicative operator *in its native host*, and what transfers
465 to the time-invariant host is the added structure, not the gating.

466 4.7. The reverse transition

467 A pre-specified grid repeats the chain’s central contrasts on the reverse
468 transition, 40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN (75 cells: frozen-all, BM3-frozen,
469 frozen-gate, S4D and S4D+gate (wm), at three levels and five seeds; Sec-
470 tion 3.1 for the data). This direction is harder for the time-invariant model:
471 on its evaluation set, predicting the majority class gives a macro-F1 of 46.8%
472 and uniform random guessing 41.5%, and S4D sits close to those levels (Ta-
473 ble 8).

474 The first two steps replicate. The fully input-independent scan exceeds
475 S4D by +26.3, +25.6 and +22.2 pp at 0, -2 and -6 dB, with intervals ex-
476 cluding zero and 5/5 seeds at every level, so by the pre-specified rule (at
477 least two of three levels) the first step is replicated on the reverse transition.
478 Removing the gate from BM3-frozen costs 18.4, 22.0 and 24.0 pp, above the
479 8 pp threshold at all three levels, with intervals excluding zero and 5/5 seeds,
480 and the gateless arm is indistinguishable from S4D (intervals include zero).
481 On the third step, which the pre-specification treats as descriptive here, the
482 width-matched graft recovers most of the gap, $R = 0.911 [0.82, 1.04]$ (0.863
483 with frozen-all in the denominator). A further pre-specified grid (15 cells)
484 adds the same-size additive control in this direction: it reaches 67.8, 72.1,
485 68.4% (final epoch) and recovers $R = 0.758 [0.66, 0.88]$. The gate arm ex-
486 ceeds it by +5.1, +1.7, +3.8 pp (only the -6 dB interval excludes zero),

Table 8: Reverse transition (40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN). Macro-F1 (%), mean over 5 seeds, best-epoch and final-epoch reporting; paired differences under final-epoch reporting with 95% t -intervals and the number of seeds (of 5) with a positive difference.

Arm (best / final)	0 dB	−2 dB	−6 dB
frozen—all	86.3 / 77.5	87.8 / 77.9	81.7 / 74.0
BM3-frozen	81.1 / 74.7	84.0 / 76.7	80.8 / 73.9
S4D+gate (wm)	79.0 / 72.9	80.5 / 73.9	78.5 / 72.2
frozen—gate	63.8 / 56.2	62.0 / 54.7	57.1 / 50.0
S4D	61.2 / 51.2	61.0 / 52.4	54.6 / 51.8
	+26.3	+25.6	+22.2
frozen—all − S4D	[21.5, 31.0] (5/5)	[21.1, 30.1] (5/5)	[18.8, 25.6] (5/5)
	+18.4	+22.0	+24.0
BM3-frozen − frozen—gate	[1.2, 35.6] (5/5)	[3.4, 40.6] (5/5)	[7.6, 40.3] (5/5)
	+5.0	+2.4	−1.8
frozen—gate − S4D	[−17.6, 27.7] (2/5)	[−19.2, 23.9] (2/5)	[−18.9, 15.3] (1/5)
	+21.7	+21.5	+20.5
S4D+gate (wm) − S4D	[15.4, 28.0] (5/5)	[17.0, 26.0] (5/5)	[15.7, 25.2] (5/5)

and the difference in recovery is $\Delta R = 0.153$ [0.06, 0.26]; under best-epoch reporting it shrinks to 0.037 [−0.1, 0.17]. By the pre-specified descriptive labels this is *unresolved*: in this direction the gate carries a small increment over the additive branch, whereas on the original transition it carried none ($\Delta R = -0.112$). Most of the reverse-direction recovery is therefore available to an additive branch of the same size, and the remaining gate increment is direction-dependent. Both ratios rest on an S4D baseline close to the trivial floor (51–52% against 46.8% for the majority class). Both statements carry the limits of this pair: the same eight bearings with their roles swapped, and a single inner-race bearing in the evaluation set.

4.8. Sensitivity to the reporting rule

The main tables use the final epoch of the 50-epoch budget (Section 3.3). The original grids recorded instead the best epoch on the evaluation condition, and their pre-specified thresholds were written for that rule. To measure how much the rule shapes the conclusions, every arm of the chain was re-run over the adjudication band with per-epoch evaluation, using the original training code and the same evaluation fingerprint: the four arms BM3-frozen, S4D, frozen—gate and S4D+gate (60 cells), the BM3 arm alongside the layered freezes (Section 4.2), and the capacity control and the

two other single-component removals, S4D-wide, frozen-conv and frozen-A (45 cells, pre-specified with a rule for changes of sign). The re-runs used a different PyTorch/Triton build from the original grids; their best-epoch values reproduce the original arm-by-condition means to within -0.6 to $+1.4$ pp across the eight re-run arms (the largest deviation is in the high-variance frozen-gate arm at -2 dB).

Table 9: Macro-F1 (%), mean over 5 seeds, under best-epoch and final-epoch reporting in the re-runs, for every arm of the chain on the adjudication band.

Arm	0 dB		-2 dB		-6 dB	
	best	final	best	final	best	final
BM3	95.6	86.9	95.8	88.3	88.6	83.3
BM3-frozen	90.6	83.1	92.3	87.3	85.7	83.1
S4D	75.9	67.6	75.1	67.5	68.6	66.7
S4D-wide	77.8	67.3	73.3	66.5	69.0	67.1
frozen-gate	77.7	74.0	73.1	71.7	61.9	60.5
frozen-conv	97.1	96.3	95.4	94.8	77.9	77.2
frozen-A	92.5	85.1	95.6	89.9	90.4	87.9
S4D+gate	80.9	71.5	79.9	74.3	69.2	66.6

Best-epoch values exceed final-epoch values by 0.6 – 10.6 pp across the 24 arm-by-level means, so best-epoch absolute levels are optimistic. The reduction is of similar size for the arms being contrasted (e.g., BM3-frozen -7.4 , -5.0 , -2.6 pp and S4D -8.3 , -7.6 , -1.9 pp at 0 , -2 , -6 dB), and the contrasts that carry the argument are correspondingly stable (Table 10).

Selectivity. The frozen block’s advantage over S4D is $+15.5$, $+19.8$ and $+16.4$ pp (and $+21.9$, $+24.0$, $+21.2$ pp for the fully frozen scan, Section 4.2), versus $+14.6$, $+17.2$ and $+17.1$ pp under best-epoch reporting in the re-run. It remains positive in 5/5, 5/5 and 4/5 seeds, with intervals excluding zero at every level. *Gate.* Removing the gate costs 9.1 , 15.6 and 22.6 pp (12.8 to 23.8 pp under best-epoch reporting in the re-run), above the 8 pp pre-specified threshold at every level; the interval at -2 dB includes zero ($[-0.8, 32.0]$ pp, 4/5 seeds) because of the large seed variance of the gateless arm (standard deviation 14.6 pp at -2 dB). *Gateless versus S4D.* Under final-epoch reporting the gateless arm is not below S4D at 0 and -2 dB (point estimates $+6.4$ and $+4.2$ pp) and is below it at -6 dB (-6.2 pp); all three intervals include zero.

Table 10: Paired differences under final-epoch reporting, in percentage points: mean, 95% t -interval over seeds, and number of seeds (of 5) with a positive difference.

Contrast	0 dB	−2 dB	−6 dB
	+15.5	+19.8	+16.4
BM3-frozen − S4D	[4.3, 26.8] (5/5)	[12.0, 27.7] (5/5)	[4.1, 28.7] (4/5)
	+9.1	+15.6	+22.6
BM3-frozen − frozen−gate	[2.9, 15.3] (5/5)	[−0.8, 32.0] (4/5)	[5.8, 39.4] (5/5)
	+6.4	+4.2	−6.2
frozen−gate − S4D	[−2.9, 15.8] (4/5)	[−14.5, 22.9] (3/5)	[−23.5, 11.2] (1/5)
	+3.9	+6.8	−0.1
S4D+gate − S4D	[−0.8, 8.6] (4/5)	[2.4, 11.1] (5/5)	[−4.0, 3.9] (2/5)

528 The statement that gate removal “falls below S4D” is therefore supported
529 only as a point-estimate observation at the deepest level, and we phrase it as
530 “to or below” elsewhere. *Graft.* The recovery ratio is 0.249, 0.341 and -0.003
531 per level and $R = 0.196$ on average (best-epoch re-run: 0.220), again in the
532 pre-specified “insufficient” bin on the point estimate. The seed-resampled
533 95% interval is $[0.09, 0.31]$ and so reaches the 0.3 boundary: with five seeds
534 the data support “far below confirmation (0.7) and probably below 0.3”, not
535 a precise value of R . The graft gain over S4D is resolved at -2 dB only
536 ($+6.8$ pp, $[2.4, 11.1]$) and is indistinguishable from zero at -6 dB.

537 *Remaining arms.* By the rule pre-specified for the 45-cell grid, no arm
538 changes sign between the two rules at any level where both values exceed
539 1 pp, so all main-table entries are reported under the final-epoch rule with
540 unchanged wording. The capacity control holds: BM3-frozen exceeds S4D-
541 wide by $+15.9$, $+20.8$, $+16.0$ pp (intervals excluding zero at all three levels),
542 and S4D-wide $-$ S4D is -0.4 , -1.0 , $+0.4$ pp (intervals including zero). For
543 the stem the final-epoch rule sharpens the regime trade-off (intervals at 0
544 and -2 dB now exclude zero) without reversing it.

545 Because the final epoch of the same evaluation condition has been in-
546 spected repeatedly during this study, last-epoch values are also not a fresh
547 confirmatory test. The reporting rule therefore changes absolute levels and
548 the strength of the gateless-versus-S4D statement, but not the ordering BM3-
549 frozen $>$ {S4D+gate, frozen−gate, S4D}, the size of the frozen block’s ad-
550 vantage over S4D, or the verdicts of the pre-specified gate and graft rules.

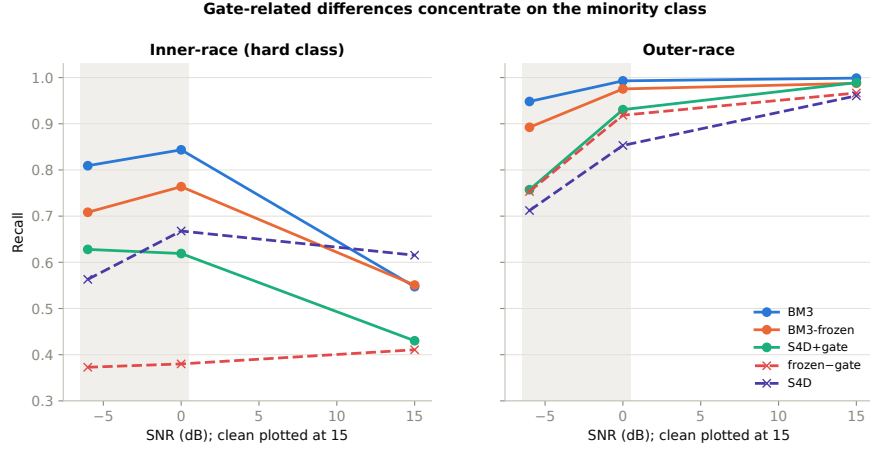


Figure 5: Per-class recall (inner-race, IR, and outer-race, OR) as a function of noise depth, split by gate presence. Gated arms lift IR recall under matched noise; gateless arms let it deteriorate.

4.9. The interaction effect concentrates on the hard class

Per-class analysis (a separate 75-cell grid at 20 epochs with best-epoch selection on the evaluation condition: 5 arms $\times$ {clean, 0, -6 dB} $\times$ 5 seeds, per-sample predictions dumped and re-verified against reported metrics cell-by-cell): the inner-race class is hard from the outset (clean recall 0.41–0.62 vs. 0.96–1.00 for outer-race). Its trajectory under noise is governed by gate presence: in all three gated arms IR recall at -6 dB exceeds its clean value (e.g., 0.55 $\rightarrow$ 0.81 for BM3), while in both gateless arms it is below its clean value (frozen-gate: 0.41 $\rightarrow$ 0.37; S4D: 0.62 $\rightarrow$ 0.56, non-monotone, rising to 0.67 at 0 dB) (Figure 5). Both ablation directions hit IR hardest—gate removal costs IR 2.3–3.8 $\times$ more F1 than OR at every level (F1 from predictions pooled over seeds), and the graft’s noise-time gains also concentrate on IR (-6 dB: +6.8 pp IR vs. +4.2 pp OR). Because macro-F1 averages the two classes, the arm-level robustness gaps of Sections 4.1–4.4 are carried substantially by the hard class. The gate-related differences are therefore class-concentrated rather than uniform, and fall on the class with the lowest recall. We did not measure classification margins, so we do not describe IR as a “low-margin” class. Because this grid is separate from the main 50-epoch grid and the IR class comes from a single training bearing (Section 3.1), these results are descriptive and do not decompose the main-grid effects.

571 5. Discussion

572 **What can be credited?** The ablation chain supports a narrow, con-
 573 ditional statement: within this Mamba-3 block, on this testbed and with
 574 noise-aware training, (i) making the scan fully input-independent—first the
 575 Δ , A , B , C projections, then the trapezoidal weight and rotation angles as
 576 well—did not remove the block’s deep-noise advantage over a time-invariant
 577 S4D, and at the deepest level the fully frozen arm matched or exceeded the
 578 original block with fewer parameters, (ii) removing the output gate removed
 579 most of that advantage—(i) and (ii) holding on the reverse transition as
 580 well—and (iii) adding the gate to S4D recovered part of the advantage only
 581 together with added width and projections, and an additive branch of the
 582 same size recovered as much, while (iv) in the Mamba-3 block an additive
 583 branch of the same size recovered none of what removing the gate cost, on
 584 both transitions. These observations are consistent with a gate benefit that is
 585 specific to the multiplicative operator in its native host. They do not identify
 586 which other component the gate depends on. Nor do they separate “input-
 587 dependence contributes nothing here” from “what it contributes is recovered
 588 by learned constants under this budget”; the two readings are indistinguish-
 589 able in our data and lead to the same attribution. Representability results
 590 for selectivity [8] and the S4D-superiority finding on clean data [4] are not
 591 contradicted by a single ablation on one testbed; our clean-data parity a-
 592 grees with the latter, and our deep-noise result suggests that in this block
 593 the output gate, rather than any input-dependent term of the scan, accounts
 594 for most of the measured advantage.

595 **Implications for backbone selection and for transplant practice.**
 596 For practitioners choosing backbones for noisy, drifting signal regimes, the
 597 results suggest that the block’s benefit should be judged as a composition,
 598 and that a gate added to a different host should not be expected to reproduce
 599 it: in our S4D host a gate built in the donor’s shape recovered no more than
 600 an additive branch of the same size on the original transition (Section 4.5)
 601 and only slightly more on the reverse one (Section 4.7), and neither design
 602 was tuned. A hypothesis for future work follows: a recomposed block guided
 603 by the component map (no conv stem, gate retained, S4D-style A) may match
 604 or exceed the original on this testbed; we leave it untested, because post-hoc
 605 recomposition after seeing the map would not carry the evidential status of
 606 the pre-specified chain.

607 **Why might gating be composition-bound?** We refrain from a mech-

608 anistic claim beyond the data, but note three compatible observations. First,
 609 the gate-related differences concentrate on the minority, lowest-recall class
 610 (Section 4.9), and matched-noise training lifts that class *only when the gate*
 611 *is present*. One hypothesis, which we did not test, is that the gate is the
 612 component through which matched-noise training improves hard-class recall,
 613 i.e., that it exploits noise-aware training rather than acting as a test-time de-
 614 noiser; we measured neither classification margins nor gate activations, and
 615 the reading is in any case bounded by the matched-SNR scope (Section 6).
 616 Second, the multiplicative form admits a per-sample suppression interpre-
 617 tation, but no gate statistics were recorded, so this remains a hypothesis
 618 rather than an observation. Third, the gateless composition falling to or
 619 below S4D at the deepest noise levels is compatible with the other compo-
 620 nents being calibrated to a gated regime (e.g., the stem may pass features
 621 that are only safe when the gate can veto them); this is the weakest of the
 622 three observations, since the gateless-S4D gap is not resolved with five seeds
 623 (Section 4.8). All are testable in future work via pairwise interaction grids
 624 and mismatched-noise protocols.

625 6. Boundaries and Limitations

626 We state the scope plainly.

- 627 1. **Testbed scope.** All claims are bounded to bearing vibration from
 628 one pair of operating conditions of one dataset (37.5 Hz/11 kN and
 629 40 Hz/10 kN; speed and load both change), binary classification, GPU
 630 execution (no edge/real-time claims). The pair is run in both direc-
 631 tions, but the reverse direction uses the same eight bearings with their
 632 roles swapped and has a single IR bearing in its evaluation set, so it is
 633 not an independent replication; the third condition of the dataset has
 634 no IR bearing. The five seeds do not resample bearings (Section 3.1).
 635 Independent bearings, multi-class settings and a second dataset are
 636 future work.
- 637 2. **Attribution level.** Credit assigned to “the rest of the composition”
 638 is by elimination within the enumerated component set; pairwise in-
 639 teraction terms are not decomposed, and the host-by-gate contrast of
 640 Section 4.4 is a four-arm estimate that cannot be assigned to a specific
 641 component.

- 642 3. **The transplant conclusion is budget-conditional.** All arms share
643 a matched 50-epoch budget; the grafted gate trains normally yet re-
644 covers little, and convergence is not robustness (Section 4.4)—but we
645 cannot exclude that a longer or graft-tuned budget would recover more.
646 Our statement is therefore limited to *this gate branch, in this host, un-*
647 *der a matched training budget.*
- 648 4. **Matched-SNR scope.** Training and evaluation share the same in-
649 jected SNR throughout (Section 3.1); “robustness” in this paper is
650 matched-noise robustness under noise-aware training, not clean-train/noisy-
651 test generalisation. The gate’s benefit may operate by exploiting matched-
652 noise training (consistent with Section 4.9) rather than by test-time de-
653 noising; attribution under mismatched-noise protocols is untested and
654 may differ.
- 655 5. **Graft dual reading.** Interaction dependence vs. transplant friction
656 (Section 4.4), the latter now subsumed under the budget condition of
657 item 3.
- 658 6. **Noise-regime coverage.** Under strong coloured noise at -10 dB a
659 TCN baseline slightly overtakes the Mamba-style block, and -10 dB
660 is near-chance for the time-invariant baseline and excluded from ad-
661 judication; the demonstrated advantage regime is matched, Gaussian-
662 dominated noise.
- 663 7. **Conv-stem dual reading.** Its clean-data cost admits both representation-
664 mistransfer and optimisation-difficulty readings, reported in place (Sec-
665 tion 4.3).
- 666 8. **Reporting rule.** Reported values are final-epoch macro-F1 under a
667 fixed 50-epoch budget. The original grids’ best-epoch rule, for which
668 the thresholds of the early grids were pre-specified, is reported as a
669 sensitivity analysis and is optimistic by 0.6–10.6 pp (Section 4.8). Ev-
670 ery arm of the chain was re-run on the three adjudication levels; the
671 $+10/+6$ /clean/ -10 dB levels and the per-class grid were not re-run
672 and are auxiliary (best epoch). No source-domain validation rule was
673 available, and because the final epoch of the same evaluation condi-
674 tion has been inspected repeatedly, final-epoch values are not a fresh
675 confirmatory test.
- 676 9. **Scope of the freezing intervention.** The chain’s first step freezes
677 the input-dependent parts of Δ , A , B , C only; the fully input-independent
678 scan is a separate grid (Section 4.2) on the three adjudication levels,
679 which was specified after the earlier grids had been seen. No equivalence

margin was pre-specified for the cost of freezing, so “no measurable cost” means the point estimates favour the frozen arms with intervals including zero, not a demonstrated equivalence. Freezing is implemented inside this Mamba-3 layer; a conclusion about input-dependence in general would need other blocks and tasks.

10. **Gate interventions.** The gate-to-additive substitution has been made in both hosts and both directions. In the Mamba-3 block the verdict is *not rescued* on both transitions, with upper interval bounds of 0.486 and 0.362 against a threshold of 0.5 (the first close to it). In S4D the width-matched arms are larger than the donor block, and on the reverse transition the gate carries a small increment over the additive branch that it does not carry on the original transition; this direction dependence is unresolved.
11. **Strength of the pre-specification evidence.** Thresholds were pre-specified internally; for all but the last three follow-up grids the supporting evidence is local file times and run-directory names, not an external time-stamped registry (Section 3.5). The process-level repeat check is a single observation.
12. **Construction checks.** For the frozen arm the state-dictionary check compares names and shapes of non-target tensors; value-level comparison was applied to the component-ablation and graft arms only.
13. **Sample size.** Every cell mean uses five seeds. The smallest attainable one-sided Wilcoxon p is 0.031, paired intervals are wide, and several secondary statements (conv stem, A parameterisation, gateless versus S4D) are not resolved at this size; conclusions rest on effect sizes that clear the pre-specified thresholds, not on significance tests, which were never adjudication criteria.

A single-repeat check of process-level non-determinism (~ 0.7 pp) and the label-hash/adjudication trail are documented in the reproducibility appendices (Appendix A–Appendix B).

7. Conclusion

In this Mamba-3 block, on one controlled drift $\times$ noise testbed with matched-noise training, freezing the input-dependent Δ , A , B , C projections costs at most about 4 pp (an estimate that five seeds leave imprecise) while 15–20 pp of advantage over a time-invariant SSM remains, and freezing the

715 trapezoidal weight and rotation angles as well—leaving no input-dependent
716 tensor in the scan—costs nothing measurable and leaves 21–24 pp of ad-
717 vantage with fewer parameters. The multiplicative output gate is the one
718 ablated component whose removal causes a large, one-signed loss (9–23 pp,
719 returning the block to roughly the level of plain S4D), yet adding the gate
720 to S4D recovers no more than the added structure does: 17–20% of the gap
721 for a narrow graft, about 30% for one built in the donor’s width, and at
722 least as much for an additive branch of the same size. Conversely, an addi-
723 tive branch of identical size and initial weights, placed where the gate was
724 in the Mamba-3 block, recovers none of the loss on either transition. The
725 gate is thus necessary as a multiplicative operator in its native host, and
726 what transfers to a time-invariant host is added structure rather than gat-
727 ing; the gate-related differences concentrate on the minority class. The first
728 two steps replicate on the reverse transition. The evidence rests on one pair
729 of operating conditions run in both directions with the same eight bearings,
730 five seeds, and final-epoch reporting under a fixed budget, with the earlier
731 evaluation-condition epoch selection retained as a sensitivity analysis. We
732 offer the three-step chain—freeze, remove, add—as a protocol for turning ar-
733 chitecture comparisons into component-level tests, with the caveat that the
734 interventions must be matched in width and capacity for the add step to be
735 conclusive, and we release our adjudication trail, including the incidents in
736 which automated adjudication departed from the pre-specified criteria.

737 **CRedit authorship contribution statement**

738 Yi Wang: Conceptualization, Methodology, Software, Validation, Formal
739 analysis, Investigation, Data curation, Writing – original draft, Writing –
740 review & editing, Visualization, Supervision, Project administration.

741 **Declaration of competing interest**

742 The authors declare that they have no known competing financial inter-
743 ests or personal relationships that could have appeared to influence the work
744 reported in this paper.

745 **Declaration of Generative AI and AI-assisted technologies in the**
746 **writing process**

747 During the preparation of this work the authors used LLM-based agents
748 to execute and cross-check pre-specified analysis and adjudication passes as
749 described in Appendix C, and to assist with manuscript drafting and figure
750 generation. After using these tools, the authors reviewed and edited the con-
751 tent as needed and take full responsibility for the content of the published
752 article. All numeric thresholds, adjudication rulings, and scientific interpre-
753 tations reported in this paper were fixed and made by the human authors,
754 as detailed in Appendix C.

755 **Data availability**

756 The code, pre-specification artifacts, and adjudication trail for this study
757 are available at <https://github.com/jeff-1980/bm3-jizi>. The reposi-
758 tory includes all follow-up grids (315 cells with per-epoch curves), their pre-
759 specifications, drivers, construction checks and decision memos and the har-
760 ness’s own imported modules, so the training entry point runs from a fresh
761 checkout.

762 **Appendix A. Fairness protocol**

763 *Appendix A.1. Evaluation-set fingerprint*

764 Every reported cell in this study is tagged with `eval_sha256`, a SHA-
765 256 hash of the raw held-out test-label byte array, computed once per run
766 and asserted identical across every arm/condition/seed cell within a grid.
767 It fingerprints *that the same held-out label vector was scored every time*, not
768 model weights or predictions. It is also the split on which the best-epoch rule
769 of Section 3.3 selects epochs. The full value shared by all 865 cells across all
770 five grids is:

771 `6c20b367522ce5db59720120f8a7cd12792c7e070c6ae84bd1985511998727de`

772 This value is recomputed independently at adjudication time (not merely
773 copied from the run log) and cross-checked row-by-row against every grid’s
774 fairness report before any comparison across arms is treated as valid.

775 *Appendix A.2. Per-arm parameter counts*

776 Table A.11 lists every arm’s parameter count, each obtained directly via
 777 `sum(p.numel() for p in model.parameters())` on the instantiated model
 778 (not estimated), cross-verified against an independently maintained archi-
 779 tecture map.

Table A.11: Per-arm parameter counts (verified against `arch_map.md` and harness source).

Arm	Parameters
BM3 (<code>bm3_kin</code> / <code>bm3_nokin</code>)	177,938
BM3-frozen (<code>bm3_frozen</code>)	112,410
S4D (<code>s4d</code>)	100,162
S4D-wide (<code>s4d_wide</code> , capacity control, $1.75\times$)	178,498
frozen-conv (<code>frozen_noconv</code>)	112,026
frozen-gate (<code>frozen_nogate</code>)	79,642
frozen-A (<code>frozen_As4d</code>)	112,410
S4D+gate (<code>s4d_plus_gate</code>)	116,802

780 *Appendix A.3. Five-batch pre-specification timeline*

781 Table A.12 lists, for each of the five pre-specified grids, the artifact that
 782 fixed its adjudication thresholds, that artifact’s creation/modification date,
 783 and the corresponding grid’s run-directory start time. In every case the
 784 prereg artifact predates the grid it governs.

785 Two honest caveats. First, the baseline batch predates the orchestration
 786 infrastructure used for the other four batches by five days; its prereg trail
 787 rests on the task card’s creation timestamp alone, not on the same automat-
 788 ed executor/reviewer machinery documented in [Appendix C](#), and we state
 789 this rather than implying a uniform process across all five batches. Second,
 790 for the component-ablation grid, two additional JSON files bearing “pre-
 791 reg” in their name exist on disk (`secondary_prereg_20260704-0708` and
 792 `secondary_deltapp_prereg_20260704-0719`) that *post-date* the grid’s start
 793 time; both are explicitly marked void in the corresponding decision artifact’s
 794 `void_preregs` field, and neither was used to adjudicate any result in this pa-
 795 per. They are the on-disk residue of the first post-hoc criterion substitution
 796 described in [Appendix C](#), retained here (rather than deleted) as part of the
 797 released adjudication trail.

Table A.12: Pre-specification predates data collection for all five grids.

Grid (cells)	Canonical prereg artifact	Prereg date	Grid start
Baseline (475)	Task-tracking-system card (<code>created:</code> 2026-06-27, mtime 15:02)	2026-06-27	2026-06-27 17:31
Frozen-selectivity (105)	<code>frozensel_prereg.json</code> (m- time 18:26:54)	2026-07-02	2026-07-02 18:27
Component ab- lation (175)	Task-tracking-system card (<code>created:</code> 2026-07-03), materialised in <code>prereg_provenance.md</code>	2026-07-03	2026-07-03 10:34
Graft (35)	<code>graft_prereg_provenance.md</code> (mtime 21:20:36)	2026-07-05	2026-07-05 22:41
Per-class (75)	Task-tracking-system card (<code>created:</code> 2026-07-06)	2026-07-06	2026-07-07 21:21

798 Appendix B. Confusion matrices and a repeat-run check

799 Appendix B.1. Full confusion-matrix set

800 Table B.13 gives the complete confusion-matrix set underlying the per-
801 class analysis of Section 4.9: 5 arms $\times$ 3 conditions (clean, 0 dB, -6 dB),
802 each summed over 5 seeds (support OR= 17,520, IR= 8,720 per cell). These
803 counts were dumped per-sample and re-verified against the reported recal-
804 l/precision/F1 metrics cell-by-cell before use.

Table B.13: Full confusion-matrix set (summed over 5 seeds; support OR=17,520, IR=8,720 per cell). Columns give true $\rightarrow$ predicted counts.

Arm	Condition	OR $\rightarrow$ OR	OR $\rightarrow$ IR	IR $\rightarrow$ OR	IR $\rightarrow$ IR
BM3	clean	17,500	20	3,948	4,772
	0 dB	17,395	125	1,364	7,356
	-6 dB	16,613	907	1,664	7,056
BM3-frozen	clean	17,303	217	3,917	4,803
	0 dB	17,089	431	2,060	6,660
	-6 dB	15,634	1,886	2,543	6,177
S4D+gate	clean	17,327	193	4,969	3,751

Arm	Condition	OR→OR	OR→IR	IR→OR	IR→IR
frozen-gate	0 dB	16,303	1,217	3,322	5,398
	−6 dB	13,267	4,253	3,244	5,476
	clean	16,935	585	5,139	3,581
	0 dB	16,092	1,428	5,406	3,314
S4D	−6 dB	13,200	4,320	5,469	3,251
	clean	16,828	692	3,354	5,366
	0 dB	14,945	2,575	2,897	5,823
	−6 dB	12,477	5,043	3,810	4,910

805 *Appendix B.2. Repeat-run check*

806 Because two independent process runs of an identical configuration are
807 not bitwise reproducible on this hardware (non-deterministic GPU kernel re-
808 ductions in the selective-scan and normalisation ops), we measured a process-
809 level noise floor directly: the same arm, condition, and seed (`bm3.frozen`,
810 clean, seed 0, 2 epochs) was run twice in separate processes with identical
811 evaluation fingerprints and identical parameter counts confirmed. The two
812 runs’ best macro-F1 differed by

$$|0.69025 - 0.69712| = 0.00687 \approx 0.687 \text{ pp},$$

813 which we report as an observed repeat difference of ~ 0.7 pp (cited in Sec-
814 tion 3.5). It is a single repeat of one arm, one condition and a 2-epoch
815 run, and therefore neither an estimate nor a bound of process variability for
816 full-length runs. It is an order of magnitude below the seed-level variabil-
817 ity observed within the deep-noise band across the curve tables underlying
818 Figures 2 and 4 (standard deviations typically 2–9 pp, e.g. BM3-frozen 5.7–
819 9.4 pp and S4D 1.7–4.5 pp at $\{0, -2, -6\}$ dB). Arm-level effects in Section 4
820 are judged against seed-level variability, not against this single repeat.

821 **Appendix C. Automation disclosure and adjudication gates**

822 *Appendix C.1. Automation disclosure*

823 The mechanical stages of the analysis pipeline—applying pre-specified nu-
824 meric thresholds to aggregated results, re-computing aggregates, re-verifying
825 evaluation fingerprints, and checking each decision against its pre-specified
826 rule—were carried out by LLM-based coding agents under an orchestration

827 harness, with every step auditable in the released trail. Manuscript drafting
828 and figure generation were also assisted by such agents (main text, Declara-
829 tion of Generative AI).

830 *Appendix C.2. Incidents*

831 Three incidents occurred in the original grids, each caught before an er-
832 roneous artefact entered a reported number: (1) on the frozen-selectivity
833 grid an automated decision substituted an unspecified criterion for the pre-
834 specified 10 pp threshold; re-applying the original rule gives the reported out-
835 come, “not the primary mechanism”; (2) on the component-ablation grid an
836 automated pass substituted an interval heuristic for the pre-specified thresh-
837 old, and post-hoc rule files were generated; the true pre-specification predated
838 the grid and was re-applied, and the post-hoc files are retained and marked
839 void; (3) on the graft grid a construction stage was falsely marked as passed;
840 the grid was re-executed from scratch under a corrected task card. Devia-
841 tions in the later follow-up grids (for example a bootstrap made deterministic
842 after first use, and rounding corrections) are recorded in the repository’s de-
843 cision memos. Full descriptions, with dates and artefact names, are given in
844 Supplementary Material S1.

845 *Appendix C.3. Human adjudication gates*

846 All numeric adjudication thresholds, every ruling on an incident, the de-
847 cision to report the incidents, and every interpretive claim in Section 5 were
848 fixed or made by the human authors. No agent could overwrite a protected
849 decision artefact, alter a pre-specified threshold or mark its own output as
850 final.

851 *Appendix C.4. Alignment with Elsevier’s generative-AI policy*

852 In line with Elsevier’s policy, AI tools are not listed as authors and the
853 human authors take full responsibility for the content. We extend the dis-
854 closure beyond the writing process to the adjudication pipeline because that
855 is where automation could affect the reported conclusions.

856 **References**

- 857 [1] A. Gu, T. Dao, [Mamba: Linear-time sequence modeling with selective](#)
858 [state spaces](#), arXiv preprint arXiv:2312.00752Also presented at COLM
859 2024 (2023).
860 URL <https://arxiv.org/abs/2312.00752>

- [2] H. Qu, L. Ning, R. An, W. Fan, T. Derr, H. Liu, X. Xu, Q. Li, A survey of mamba, arXiv preprint arXiv:2408.01129 (2024).
- [3] B. N. Patro, V. S. Agneeswaran, Mamba-360: Survey of state space models as transformer alternative for long sequence modelling: Methods, applications, and challenges, arXiv preprint arXiv:2404.16112 (2024).
- [4] H. Saadatmand, G. I. Webb, H. Rezatofighi, M. Salehi, [A simple state space model excels at multivariate time series classification](https://arxiv.org/abs/2605.27406), arXiv preprint arXiv:2605.27406 Systematic S4D-vs-Mamba-family comparison, 59 datasets (MONSTER + UEA); submitted 2026-05-07 (2026). URL <https://arxiv.org/abs/2605.27406>
- [5] A. Yu, N. B. Erichson, [Block-biased mamba for long-range sequence processing](https://arxiv.org/abs/2505.09022), arXiv preprint arXiv:2505.09022 (2025). URL <https://arxiv.org/abs/2505.09022>
- [6] A. Gu, A. Gupta, K. Goel, C. Ré, [On the parameterization and initialization of diagonal state space models](https://arxiv.org/abs/2206.11893), in: Advances in Neural Information Processing Systems (NeurIPS), Vol. 35, 2022. URL <https://arxiv.org/abs/2206.11893>
- [7] A. Lahoti, K. Y. Li, B. Chen, C. Wang, A. Bick, J. Z. Kolter, T. Dao, A. Gu, [Mamba-3: Improved sequence modeling using state space principles](https://arxiv.org/abs/2603.15569), in: International Conference on Learning Representations (ICLR), 2026, arXiv:2603.15569. URL <https://arxiv.org/abs/2603.15569>
- [8] N. Huang, M. Sarabia, A. Moudgil, P. Rodríguez, L. Zappella, F. Danieli, [Understanding input selectivity in mamba: Impact on approximation power, memorization, and associative recall capacity](https://arxiv.org/abs/2506.11891), arXiv preprint arXiv:2506.11891 (2025). URL <https://arxiv.org/abs/2506.11891>
- [9] M. Shandirasegaran, H. Li, S. Zhang, M. Wang, S. Zhang, A theoretical analysis of mamba’s training dynamics: Filtering relevant features for generalization in state space models, arXiv preprint arXiv:2602.12499 (2026).
- [10] H. Yi, D. Li, Z. Lu, Y. Jin, H. Duan, L. Hou, F. Z. Duraihem, E. M. Awwad, N. A. Saeed, [Vibrmamba: A lightweight mamba based fault](https://arxiv.org/abs/2602.12499)

- 894 diagnosis of rotating machinery using vibration signal, Measurement-
 895 Published online 2025-02-01 (2025).
 896 URL [https://www.sciencedirect.com/science/article/abs/pii/](https://www.sciencedirect.com/science/article/abs/pii/S0263224125002404)
 897 [S0263224125002404](https://www.sciencedirect.com/science/article/abs/pii/S0263224125002404)
- 898 [11] D.-T. Nguyen, V.-Q.-V. Nguyen, T.-T. Tran, V.-T. Pham, [Mlfork: Bear-](#)
 899 [ing fault diagnosis via mamba-powered few-shot learning model with](#)
 900 [multi-level architecture enhanced by spatial-wise and channel-wise local](#)
 901 [vector attention](#), Neurocomputing 656 (2025) 131518.
 902 URL [https://www.sciencedirect.com/science/article/abs/pii/](https://www.sciencedirect.com/science/article/abs/pii/S0925231225021903)
 903 [S0925231225021903](https://www.sciencedirect.com/science/article/abs/pii/S0925231225021903)
- 904 [12] P. Wang, Y. Song, X. Wang, Q. Xiang, [Md-bimamba: An aero-engine](#)
 905 [inter-shaft bearing fault diagnosis method based on mamba with modal](#)
 906 [decomposition and bidirectional features fusion strategy](#), Measurement
 907 (2025).
 908 URL [https://www.sciencedirect.com/science/article/abs/pii/](https://www.sciencedirect.com/science/article/abs/pii/S026322412401755X)
 909 [S026322412401755X](https://www.sciencedirect.com/science/article/abs/pii/S026322412401755X)
- 910 [13] J. Xu, Y. Zhao, W. Bao, C. Hao, [Fault diagnosis of motor bearing in](#)
 911 [complex scenarios based on mamba and indicative contrastive learning](#),
 912 Engineering Applications of Artificial Intelligence (2025).
 913 URL [https://www.sciencedirect.com/science/article/abs/pii/](https://www.sciencedirect.com/science/article/abs/pii/S0952197625002167)
 914 [S0952197625002167](https://www.sciencedirect.com/science/article/abs/pii/S0952197625002167)
- 915 [14] S. Lai, T.-H. Cheung, K.-C. Fung, K. Xue, J. Zhao, H. L. Goshu, Z. Lyu,
 916 K.-M. Lam, [Fault diagnosis of rolling bearings using denoising multi-](#)
 917 [channel mixture of cnn and mamba-enhanced adaptive self-attention](#)
 918 [lstm](#), Sensors 25 (21) (2025) 6652. doi:10.3390/s25216652.
 919 URL <https://www.mdpi.com/1424-8220/25/21/6652>
- 920 [15] B. Wang, Y. Lei, N. Li, N. Li, A hybrid prognostics approach for estimat-
 921 ing remaining useful life of rolling element bearings, IEEE Transactions
 922 on Reliability 69 (1) (2020) 401–412.